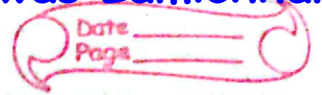


The Application Layer



- ↳ It is closest to the user - meaning the apps you use (like browsers, Gmail, WhatsApp) all work at this layer
- ↳ It acts as an interface between user and application.
- ↳ This layer is the 'face' of n/w that user interacts with

* Functions of Application Layer

① Data Representation

- ↳ ensures that data is in a readable format for both the sender and receiver, including data translation, compression, and encryption.

② Network Service Access:

- ↳ It provides applications with access to network service, enabling communication over a network

③ Application Protocols:

- ↳ It supports protocols like HTTP, FTP, SMTP, & DNS that enable communication between applications.

④ Session Management:

- ↳ It manages establishment, maintenance, and termination of communication sessions between applications

* Protocols :

① DNS (Domain Name System)

↳ helps to convert website names (like google.com) into IP addresses (like 140.250.190.78), which computers use to identify each other

• Why DNS is needed?

- ↳ Humans remember names, not numbers
- ↳ IP addresses can change
- ↳ No need to update every time IP changes
- ↳ Helps find websites fast and globally.

• Features

- ↳ distributed system, hierarchical, reliable & fast, easy to use, scalable.

• Types of DNS

- Generic domain (gTLD)
 - ↳ used for global websites.
 - ↳ .com, .org, .net, .edu, .gov, etc

- Country Domain (ccTLD)

- ↳ based on country codes (2 letters)
- ↳ .in (India), .np (Nepal), .us (USA), etc

- Inverse Domain

- ↳ used to map IP address back to domain name.

* How DNS works?

Lets say you want to visit a website:

Step 1: Your computer checks cache

IF yes, it uses that

IF no, it continues to next step.

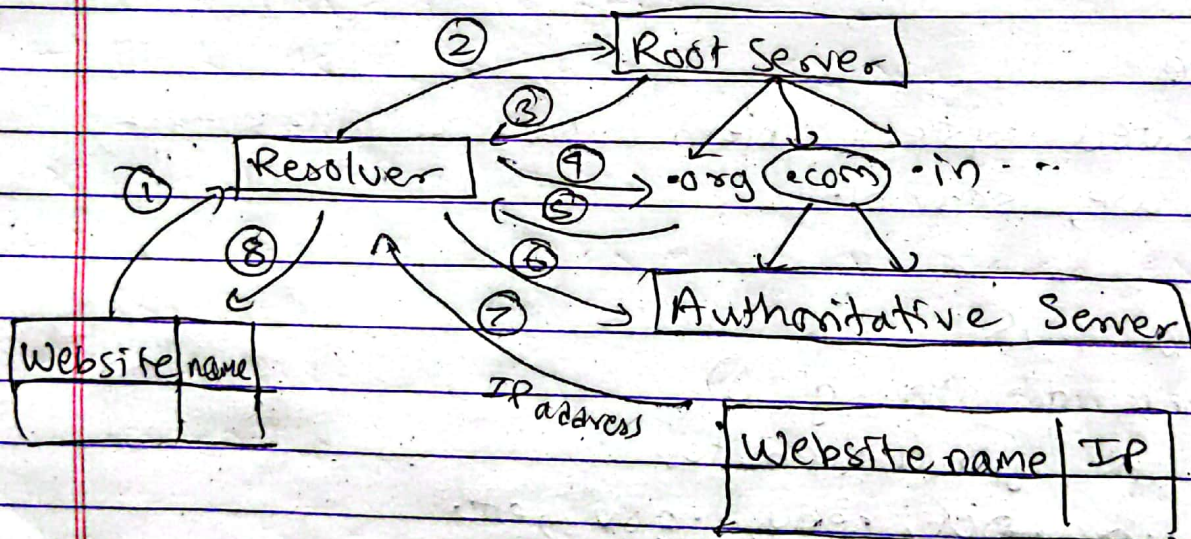
Step 2: Ask the DNS Resolver.

Step 3: Go to the Root Server

Step 4: Go to TLD Server TLD → Top Level Domain

Step 5: Go to Authoritative Name Server

Step 6: Send the IP Back



* DNS Query Types:

1) Recursive Query

↳ The resolver takes the full responsibility to go through:

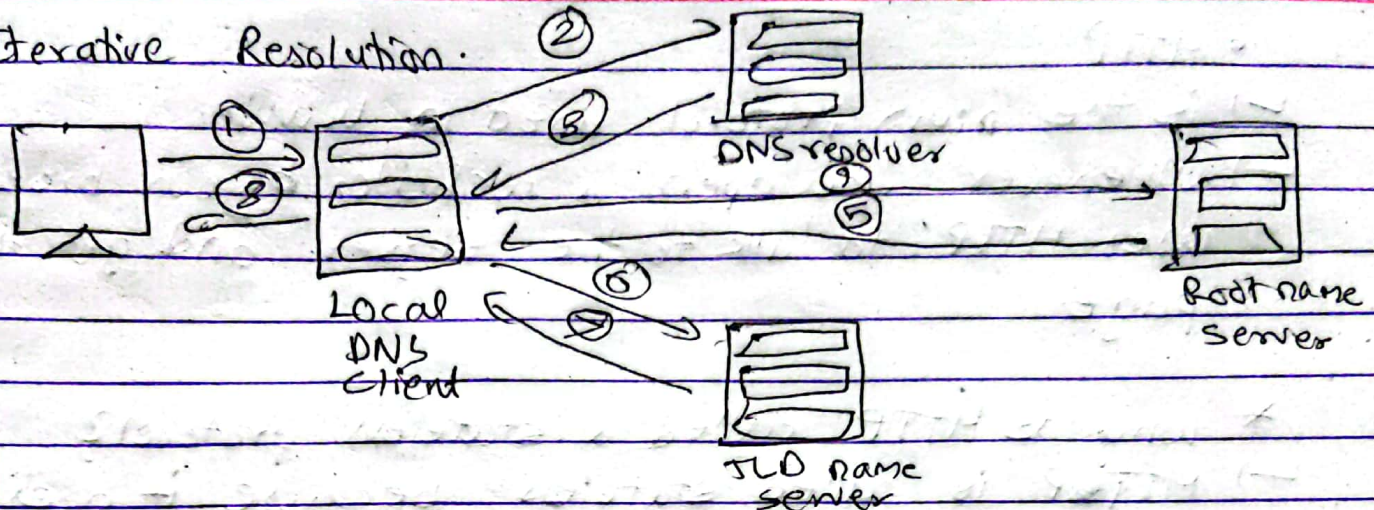
↳ Root server

↳ TLD Server

↳ Authoritative Server

↳ Then it returns final IP address to your system.

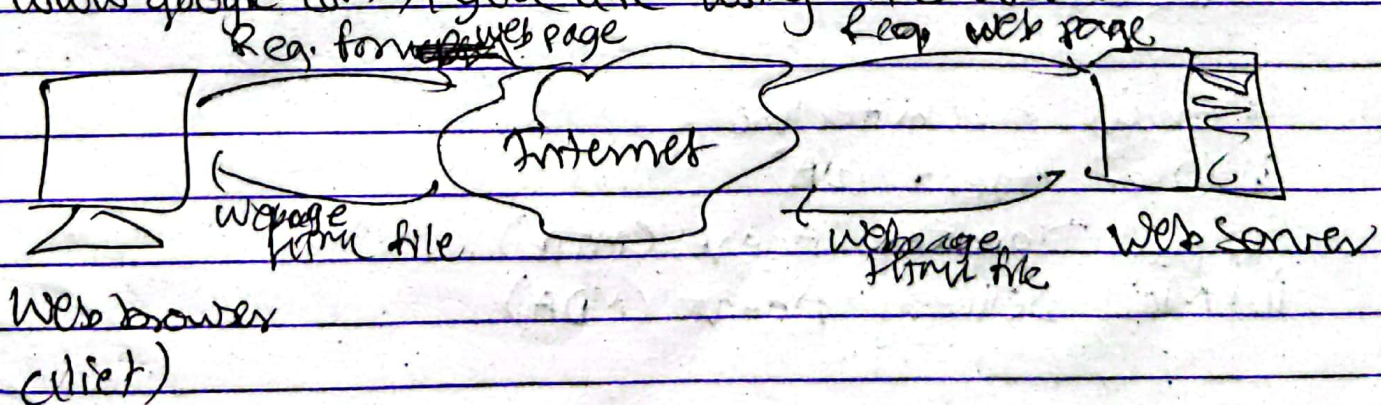
2) Iterative Resolution.



- ↳ Client does the step-by-step job
- ↳ Your computer goes to each server one-by-one to find the final answer if resolver can't find it.

* WWW (World Wide Web):

- ↳ Is a system of webpages that you can access using the Internet.
- ↳ When you open a browser like Chrome or Firefox and type in a website address (like www.google.com), you are using the WWW.



• HTTP

↳ is the main protocol used in WWW.

↳ Whenever you open a website, your browser uses HTTP to talk to the server and get the page.

* Why is HTTP called a stateless protocol?

↳ HTTP is called stateless because it does not remember anything from the previous request.

↳ Every time you send a new request to server, the server ~~don't~~ doesn't know who you are or what you did before.

↳ It treats each request as new.

* E-mail:

↳ is a method of exchanging messages over the internet.

↳ Just like sending a letter, but digitally & instantly.

• Email Architecture

i) User Agent (UA)

ii) Mail Transfer Agent (MTA)

iii) Mail Delivery Agent (MDA)

• Protocol:

• SMTP: Sends email to server

• POP3: Downloads email to your pc then deletes it from server

- IMAP: Lets you view emails without deleting from the server. (used in Mobile, Gmail)
- MIME: allows emails to send images, documents, videos.
- FTP: uploading ~~or~~ downloading files in drive, etc.

* TELNET (Telecommunication Protocol):

- ↳ is a network protocol ^{used} to remotely login to another computer over the internet or a local network.
- ↳ TELNET is not secure.

* SSH (Secure Shell)

- ↳ is a secure network protocol used to remotely access and control another computer or server over a network.
- ↳ used by network administrators, developers & IT professionals.
- ↳ SSH encrypts all data you send & receive.